

1	2	3	4	5	6
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TOP PLAN

looking down through the stack; hidden lines removed



ISOMETRIC - EXPLODED STACK

hidden lines removed; NTS: exploded for clarity – assembled device has zero gaps

C

vertical NTS; red = interface under test (Δ induced); (a) and (b) are different samples



DETAIL A — Bi_2Se_3 SURFACE STATE (ISO)

momentum space – not a physical part



DETAIL B - PHASE WINDING, 45° BILAYER (ISO)

momentum space – phase spirals 4π per Fermi loop

E

NOTES:

1. LINE CONVENTIONS: HEAVY = OUTLINE/SILHOUETTE, LIGHT = VISIBLE EDGE, HIDDEN (SHORT DASH) = BURIED MATERIAL (GATE METAL), PHANTOM (LONG-SHORT-SHORT) = NON-MATERIAL REGION (THE FIELD-DEFINED CHANNEL), DASH-DOT = CENTERLINE. NO FILLS.
2. PROVENANCE: [M] COMPUTED FROM THE VALIDATED BdG MODEL (tvqpu.lattice, 17-CHECK GATE); [L] LITERATURE; [A] ASSUMED – REPRESENTATIVE ONLY. NO DRAWN THICKNESS ENTERS THE MODEL.
3. ELEVATIONS AND ISO ARE VERTICALLY NTS (LAYERS SPAN 1.5 nm TO 0.5 nm); ISO IS EXPLODED FOR CLARITY – THE ASSEMBLED DEVICE HAS ZERO GAPS.
4. DETAILS A AND B ARE MOMENTUM-SPACE GEOMETRY. THE SPIRALS ARE THE SPIN TEXTURE AND THE ORDER-PARAMETER PHASE – NOT PHYSICAL COILS. NO REAL-SPACE SPIRAL STRUCTURE EXISTS ANYWHERE IN THIS DEVICE.
5. (a) AND (b) ARE DIFFERENT SAMPLES BY DESIGN: SAME INTERFACE UNDER TEST (RED), STACKING ORDER DIFFERS – (b) MUST BE PROBED, (a) MUST BE GATED.
6. BI-2212 DEGRADES IN AIR WITHIN MINUTES – INERT/CRYOGENIC ASSEMBLY; FABRICATION ORDER OF (b) IS AN OPEN QUESTION TO COLLABORATORS.
7. SOURCE: GRADY 2026 – github.com/VariableYT/Twisted-Cuprate-Majorana.

KEY

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|---|-------------------------------------|
|  | OUTLINE / SILHOUETTE (HEAVY) |
|  | VISIBLE EDGE (LIGHT) |
| | GATE ELECTRODE, BURIED (HIDDEN) |
|  | CHANNEL, FIELD-DEFINED (PHANTOM) |
|  | CENTERLINE (DASH-DOT) |
|  | INTERFACE UNDER TEST (Δ) |
|  | MAJORANA MODES γ_1, γ_2 |
- [M] MODEL [L] LITERATURE [A] ASSUMED



VS

VARIABLE SYSTEMS

EL DORADO HILLS, CA

GEOMETRY SHEET – NOT FABRICATION DATA

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